Project Design Phase
Solution Architecture

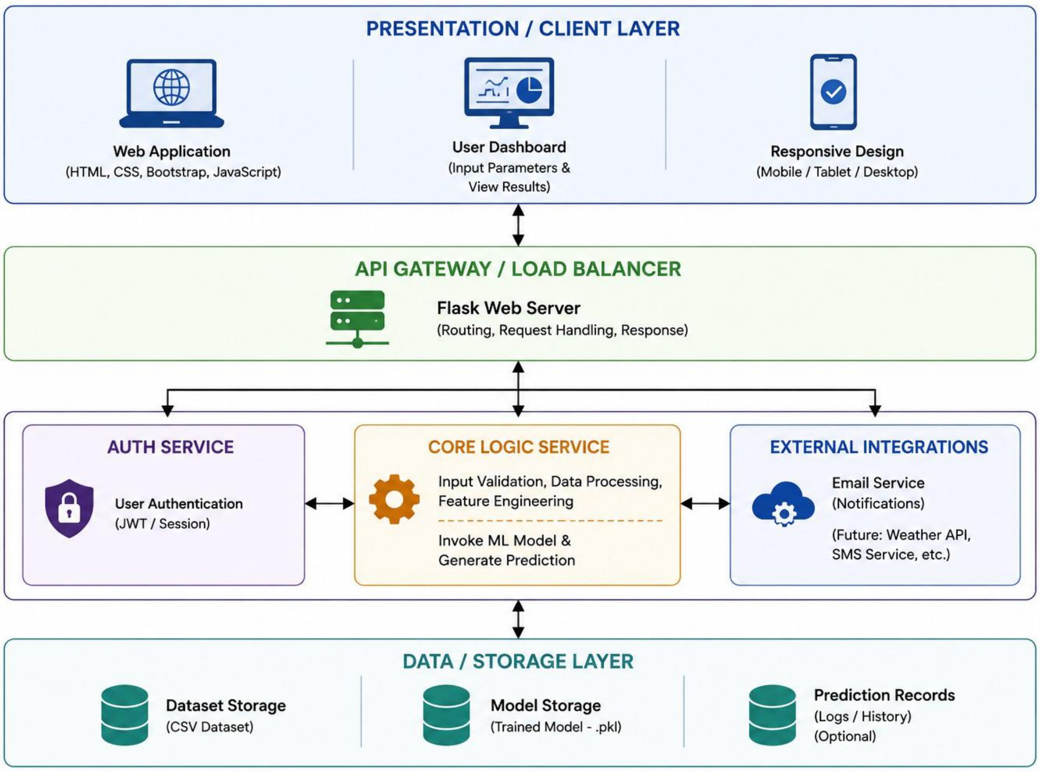
Date	10 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:



Component Description Table:

Component Name	Description/Role	Technologies Used
Presentation Layer	Provides a responsive web interface for entering soil and environmental parameters and displaying crop recommendations.	HTML5, CSS3, Bootstrap 5, JavaScript
API Gateway / Application Server	Handles HTTP requests, routes user input to backend modules, validates requests, and returns prediction results.	Flask, Python
Core Logic Service	Performs input preprocessing, loads the trained ML model, predicts the most suitable crop, and generates recommendations.	Scikit-learn, Random Forest, Pandas, NumPy, Pickle
Data / Storage Layer	Stores the agricultural dataset used for training and the trained machine learning model for prediction.	CSV Dataset, model.pkl